Multiple Liear Regression

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Multiple Linear Regression

The multiple linear regression takes the form

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_d x_d + \xi = \vec{x} \cdot \vec{\beta} + \xi$$

with $\{\beta_i\}_{i=0}^{d} \in \mathbb{R}$ constants or parameters of the model. In vector notation, $\vec{\beta} \in \mathbb{R}^{d+1}$,

$$\vec{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_d \end{pmatrix}; \quad \vec{x} = \begin{pmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}$$

For n data points, in matrix algebra notation, we can write $y = X\vec{\beta} + \xi$ where $X \in \mathcal{M}_{n\times(d+1)}$ and $y \in \mathbb{R}^{d+1}$ with

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1d} \\ 1 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}; \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad \xi = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{pmatrix}$$

We fit the n data points with the objective to minimize the loss function, mean squared error

$$MSE(\vec{\beta}) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f_{\vec{\beta}}(\vec{x}_i)\right)^2 = \frac{1}{n}\left|\vec{y} - X\vec{\beta}\right|^2$$

Ordinary Least Square Method

The `scikit-learn` library uses ***Ordinary Least Squares (OLS)*** method to find the parameters. This method is good for a simple and relatively smaller dataset. Here is a short note on this method. However, when the dimension is very high and the dataset is bigger, `scikit-learn` uses another method called ***Stochastic Gradient Descent*** for optimization which is discussed in the next section.

The goal of OLS is to find the parameter vector $\hat{\beta}$ that minimizes the sum of squared errors (SSE) between the observed target values y and the predicted values $\hat{y}$:

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}(y_i - X_i\beta)^2$$

This can be expressed in matrix form as:

$$\text{SSE} = (y - X\beta)^T(y - X\beta)$$

To minimize the SSE, let's first expand the expression:

$$\begin{aligned} \text{SSE} &= (y - X\beta)^T(y - X\beta) && (1) \\ &= (y^T - \beta^T X^T)(y - X\beta) && (2) \\ &= y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X\beta && (3) \end{aligned}$$

Since $\beta^T X^T y$ is a scalar (a 1x1 matrix), it is equal to its transpose. That is

$$
\begin{aligned}
\beta^T X^T y &= \left(\beta^T X^T y\right)^T \\
&= \left((\beta^T X^T) y\right)^T \\
&= y^T (\beta^T X^T)^T \\
&= y^T (\beta^T X^T)^T \\
&= y^T \left(X^T\right)^T \left(\beta^T\right)^T \\
&= y^T X \beta
\end{aligned}
$$

and therefore,

$$
\text{SSE} = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta
$$

To find the minimum of the SSE, we take the derivative with respect to β and set it to zero:

$$
\frac{\partial \text{SSE}}{\partial \beta} = -2X^T y + 2X^T X \beta = 0
$$

Now, solve for β:

$$
X^T X \beta = X^T y
$$

To isolate β, we multiply both sides by $(X^T X)^{-1}$ (assuming $X^T X$ is invertible):

$$
\beta = (X^T X)^{-1} X^T y
$$

The vector $\hat{\beta} = (X^T X)^{-1} X^T y$ gives the estimated coefficients that minimize the sum of squared errors between the observed target values y and the predicted values $\hat{y} = X\hat{\beta}$. This method is exact and works well when $X^T X$ is invertible and the dataset size is manageable. This method is very efficient for small to medium-sized datasets but can become computationally expensive for very large datasets due to the inversion of the matrix $X^T X$.

Iterative Method

Gradient Descent

GIF Credit: gbhat.com
Gradient Descent is an optimization algorithm used to minimize the cost function. The cost function $f(\beta)$ measures how well a model with parameters β fits the data. The goal is to find the values of β that minimize this cost function. In terms of the iterative method, we want to find β_{k+1} and β_k such that $f(\beta_{k+1}) < f(\beta_k)$. For a small change in β, we can approximate $f(\beta)$ using Taylor series expansion

$$f(\beta_{k+1}) = f(\beta_k + \Delta\beta_k) \approx f(\beta_k) + \nabla f(\beta_k)^T \Delta\beta_k + \text{higher-order terms}$$

The update rule for vanilla gradient descent is given by:

$$\beta_{k+1} = \beta_k - \eta \nabla f(\beta_k)$$

Where:

- β_k is the current estimate of the parameters at iteration k.
- η is the learning rate, a small positive scalar that controls the step size.
- $\nabla f(\beta_k)$ is the gradient of the cost function f with respect to β at the current point β_k.

The update rule comes from the idea of moving the parameter vector β in the direction that decreases the cost function the most.

1. **Gradient**: The gradient $\nabla f(\beta_k)$ represents the direction and magnitude of the steepest ascent of the function f at the point β_k. Since we want to minimize the function, we move in the opposite direction of the gradient.

2. **Step Size**: The term $\eta \nabla f(\beta_k)$ scales the gradient by the learning rate η, determining how far we move in that direction. If η is too large, the algorithm may overshoot the minimum; if it's too small, the convergence will be slow.

3. **Iterative Update**: Starting from an initial guess β_0, we repeatedly apply the update rule until the algorithm converges, meaning that the changes in β_k become negligible, and β_k is close to the optimal value β^*.

Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent is a variation of the vanilla gradient descent. Instead of computing the gradient using the entire dataset, SGD updates the parameters using only a single data point or a small batch of data points at each iteration. The later one we call it mini batch SGD.

Suppose our cost function is defined as the average over a dataset of size n:

$$f(\beta) = \frac{1}{n} \sum_{i=1}^{n} f_i(\beta)$$

Where $f_i(\beta)$ represents the contribution of the i-th data point to the total cost function. The gradient of the cost function with respect to β is:

$$\nabla f(\beta) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(\beta)$$

Vanilla gradient descent would update the parameters as:

$$\beta_{k+1} = \beta_k - \eta \nabla f(\beta_k)$$

Instead of using the entire dataset to compute the gradient, SGD approximates the gradient by using only a single data point (or a small batch). The update rule for SGD is:

$$\beta_{k+1} = \beta_k - \eta \nabla f_{i_k}(\beta_k)$$

Where:

- i_k is the index of a randomly selected data point at iteration k.
- $\nabla f_{i_k}(\beta_k)$ is the gradient of the cost function with respect to the parameter β_k, evaluated only at the data point indexed by i_k.

Python Execution

Synthetic Data

```
import numpy as np
from sklearn.linear_model import LinearRegression
X=np.random.randn(1000,2)
y=3*X[:,0]+2*X[:,1]+1+np.random.randn(1000)
```

So for this project, our known relationship is $y = 1 + 3x_1 + 2x_2 + \xi$.

Fit the data: Using scikit-learn Library

```
mlr=LinearRegression()
mlr.fit(X,y)
coefficients=mlr.coef_.tolist()
slope=mlr.intercept_.tolist()
```

So the model parameters: slope $\beta_0 = 1.0119$ and coefficients $\beta_1 = 3.0143$, and $\beta_2 = 2.034$

Fit the data: Using Custom Library OLS

First we create our custom `NewLinearRegression` using the OLS formula above and save this `python` class as `mlreg.py`

```
import numpy as np


class NewLinearRegression:
    def __init__(self) -> None:
        self.beta = None

    def fit(self, X, y):
        X = np.concatenate([np.ones((len(X), 1)), X], axis=1)
        X_transpose_X = np.dot(X.transpose(), X)
        X_transpose_X_inverse = np.linalg.inv(X_transpose_X)
        X_transpose_y = np.dot(X.transpose(), y)
        self.beta = np.dot(X_transpose_X_inverse, X_transpose_y)
```

```
    def predict(self, X):
        X = np.concatenate([np.ones((len(X), 1)), X], axis=1)
        return np.dot(X, self.beta)

    def coeff_(self):
        return self.beta[1:].tolist()

    def interceptt_(self):
        return self.beta[0].tolist()
```

Now it's time to use the new class

```
from mlreg import NewLinearRegression
mlr1 = NewLinearRegression()
mlr1.fit(X,y)
coefficients1=mlr1.coeff_()
slope1=mlr1.interceptt_()
```

So the model parameters: slope $\beta_0 = 1.0119$ and coefficients $\beta_1 = 3.0143$, and $\beta_2 = 2.034$

Fit the data: Using Gradient Descent

We create the class

```
class GDLinearRegression:
    def __init__(self, learning_rate=0.01, number_of_iteration=1000) -> None:
        self.learning_rate = learning_rate
        self.number_of_iteration = number_of_iteration
        self.weights = None
        self.bias = None

    def fit(self, X, y):
        num_of_samples, num_of_features = X.shape
        self.weights = np.zeros(num_of_features)
        self.bias = 0

        for _ in range(self.number_of_iteration):
            y_predicted = np.dot(X, self.weights) + self.bias

            d_weights = (1 / num_of_samples) * np.dot(X.T, (y_predicted - y))
```

```
            d_bias = (1 / num_of_samples) * np.sum(y_predicted - y)

            self.weights -= self.learning_rate * d_weights
            self.bias -= self.learning_rate * d_bias

    def predict(self, X):
        y_predicted = np.dot(X, self.weights) + self.bias
        return y_predicted

    def coefff_(self):
        return self.weights.tolist()

    def intercepttt_(self):
        return self.bias
```

Now we use this similarly as before,

```
from mlreg import GDLinearRegression
mlr2= GDLinearRegression(learning_rate=0.008)
mlr2.fit(X,y)
coefficients2=mlr2.coefff_()
slope2=mlr2.intercepttt_()
```

So the model parameters: slope $\beta_0 = 1.0114$ and coefficients $\beta_1 = 3.0129$, and $\beta_2 = 2.0334$

Fit the data: Using Stochastic Gradient Descent

First we define the class

```
class SGDLinearRegression:
    def __init__(self, learning_rate=0.01, num_iterations=1000, batch_size=1) -> None:
        self.learning_rate = learning_rate
        self.num_iterations = num_iterations
        self.batch_size = batch_size
        self.theta = None
        self.mse_list = None  # Initialize mse_list as an instance attribute

    def _loss_function(self, X, y, beta):
        num_samples = len(y)
        y_predicted = X.dot(beta)
        mse = (1/num_samples) * np.sum(np.square(y_predicted - y))
```

```
        return mse

    def _gradient_function(self, X, y, beta):
        num_samples = len(y)
        y_predicted = X.dot(beta)
        grad = (1/num_samples) * X.T.dot(y_predicted - y)
        return grad

    def fit(self, X, y):
        # Adding the intercept term (bias) as a column of ones
        X = np.concatenate([np.ones((len(X), 1)), X], axis=1)
        num_features = X.shape[1]
        self.theta = np.zeros((num_features, 1))

        self.mse_list = np.zeros(self.num_iterations)  # Initialize mse_list

        for i in range(self.num_iterations):
            # Randomly select a batch of data points
            indices = np.random.choice(
                len(y), size=self.batch_size, replace=False)
            X_i = X[indices]
            y_i = y[indices].reshape(-1, 1)

            # Compute the gradient and update the weights
            gradient = self._gradient_function(X_i, y_i, self.theta)
            self.theta = self.theta - self.learning_rate * gradient

            # Calculate loss for the entire dataset (optional)
            self.mse_list[i] = self._loss_function(X, y, self.theta)

        return self.theta, self.mse_list

    def predict(self, X):
        # Adding the intercept term (bias) as a column of ones
        X = np.concatenate([np.ones((len(X), 1)), X], axis=1)
        return X.dot(self.theta)

    def coef_(self):
        # Return the coefficients (excluding the intercept term)
        return self.theta[1:].flatten().tolist()

    def intercept_(self):
```

```
        # Return the intercept term
        return self.theta[0].item()

    def mse_losses(self):
        # Return the mse_list
        return self.mse_list.tolist()
```

Now

```
import matplotlib.pyplot as plt
from mlreg import SGDLinearRegression
mlr3=SGDLinearRegression(learning_rate=0.01, num_iterations=1000, batch_size=10)
theta, _ = mlr3.fit(X, y)
```

So the model parameters: slope β_0 = array([1.01399583]) and coefficients β_1 = array([3.01731174]), and β_2 = array([2.03142459])

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